Day 7: Follow the rule, then notice more

Name _____ Date _____

I can follow a rule to make a number pattern. I can notice more and say why.

Warm-up.

Find all the factor pairs of 65.	**Is 74 a multiple of 4? Show how you know.**	**Is 95 a multiple of 5? Show how you know.**

Example 1. Start at 3. The rule is add 4. Write the first eight numbers. What is true that the rule did not say? Why will it keep going?

1st	**2nd**	**3rd**	**4th**	**5th**	**6th**	**7th**	**8th**

I notice _____

It will keep going because _____

_____

Fix it 1. Rosa followed the rule start at 3, add 4. She wrote 4, 8, 12, 16.

What went wrong: _____

Right: _____ _____ _____ _____

Tapes. Write on your recording page, patterns 1 and 2.

Tape 1. Start at 1. Add 3. The tape shows 1, 4, 7. Write the next five numbers. Mark one thing the rule did not say. Why will it keep going?

Tape 5 (pattern 2 on your recording page). Start at 3. Add 5. The tape shows 3, 8, 13. Write the next five numbers. Mark one thing the rule did not say. Why will it keep going?

Fix it 2. Theo used the rule start at 1, add 3: 1, 4, 7, 10. He says: "I notice it adds 3."

What went wrong: _____

Something the rule did not say: _____

_____

Fix it 3. Lena used the rule start at 1, add 3. She noticed odd and even take turns. She says: "It keeps going because 13, 16 and 19 do it too."

What went wrong: _____

A reason: _____

_____

On your own. Write on your recording page, patterns 3 and 4.

Pattern 3. Start at 2. The rule is add 3. Write the first eight numbers. Write one thing the rule did not say. Why will it keep going?

Pattern 4. Start at 3. The rule is multiply by 2. Write the first eight numbers. Write one thing the rule did not say. Why will it keep going?

Challenge. Make a rule whose numbers take turns ending in 4 and 9. Write the start, the rule and six numbers. Why does it keep going?

Start _____ Rule _____

_____ _____ _____ _____ _____ _____

Why: _____

_____

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